

Coding & Solution

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Emotion Detection and Learning Support Engine has been developed using Python, Streamlit, TensorFlow, and NLP techniques. The application follows a modular architecture for improved readability, maintainability, and reusability. The solution successfully addresses the problem of identifying learners' emotions from textual input and providing personalized learning guidance through AI-based emotion detection.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/dwarampudilasya-rgb/Emotion-Detection-Learning-Support-Engine
Programming Language	Python
Framework(s) Used	Streamlit, TensorFlow, Keras, Pandas, NumPy, Scikit-learn, Matplotlib, Google Gemini API
Key Features Implemented	Emotion detection (BiLSTM), NLP preprocessing, Prediction confidence, Primary/secondary emotion detection, Personalized guidance, Prediction logging, Analytics dashboard, Streamlit UI, GitHub version control
Pending / Incomplete Features	None
Setup / Run Instructions	1. Clone repository. 2. Install deps: pip install -r requirements.txt 3. Run: streamlit run App/app.py 4. Open localhost URL, enter text, and view results.

Code Quality Checklist

S.N	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: The project has been developed following good software engineering practices with a modular architecture, reusable components, organized project structure, and clear documentation. The solution successfully demonstrates AI-based emotion detection, personalized learning guidance, prediction logging, and analytics visualization.